

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III (NEP) Examination 2024-25

CHEMISTRY**Paper No. CHEMJ21 / CHEMN21 : Basic Concepts of Chemistry-II**

Time: 3 Hours

Maximum Marks: 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

Attempt five questions including Question 1 which is compulsory. Attempt at least one question from Questions 2 and 3. The figures in the right-hand margin indicate marks.

1. Attempt all of the following:

- (a) "Carbon possesses only two unpaired spins in its ground state yet it forms four bonds." Why? [2]
- (b) Comment upon stability of Be₂ as per molecular orbital theory. [2]
- (c) "The H-C-F bond angle is slightly smaller than H-C-H in CH₃F." Explain this observation in terms of Bent's rule. [1]
- (d) Explain why the phenols are stronger acids than alcohols but weaker than carboxylic acids. [2]
- (e) Why does the fully-eclipsed Newman projection of n-butane have maximum torsional strain? [2]
- (f) Why does only the E2 elimination show the kinetic isotope effect while E1 does not? [2]
- (g) In Williamson's ether synthesis, why is a primary alkyl halide used instead of a secondary or tertiary one? [2]
- (h) How many transition states exist in the S_N1 reaction? [1]

2. Answer the following:

- (a) Explain in brief the Valence Bond Theory of metallic bonding through a suitable example. [4]
- (b) "The boiling point of HF is much higher than that of HCl, HBr, and HI, while in aqueous state it is the poorest acid among all." Why? [4]
- (c) Discuss the structure and bonding of either Methyl Lithium or Diborane. [6]
- (d) Draw the crystal structure of Rock salt (NaCl) and mention the coordination numbers of sodium and chloride ions. [4]
- (e) Why is the Si-F bond stronger than the corresponding C-F bond? [2]

3. Answer the following:

- (a) Let A and B be two atoms with wave functions for 1s orbitals as ψ_A and ψ_B , respectively. Write the molecular orbital wave functions using these atomic orbital wave functions ψ_A and ψ_B . Write expressions using molecular orbitals for the probability of finding the electron density in a unit volume and interpret the meaning of each term in the expressions. [6]
- (b) "The compounds trimethylamine (CH₃)₃N and trimethylsilylamine (Me₃Si)₃N have similar formulae but have pyramidal and trigonal planar structures, respectively." Why? [4]
- (c) Explain in brief the Molecular Orbital Theory of metallic bonding through a suitable example. [4]
- (d) Comment upon the variation of bond angles in CH₄, NH₃, and H₂O in the light of VSEPR theory. [4]
- (e) Explain the defects noted in extending homonuclear diatomic molecular orbital concepts to the heteronuclear diatomic molecule CO using an energy level diagram. [2]

4. Answer the following:

- (a) Write down the product(s) of the following reactions: [8]
- (i) $\text{H}_3\text{C} - \text{C} \equiv \text{CH} \xrightarrow[\text{dil. H}_2\text{SO}_4]{\text{HgSO}_4} ?$
- (ii) Cyclohexene $\xrightarrow[2. \text{H}^+]{1. \text{MCPBA}} ?$
- (iii) $n \text{H}_3\text{C} - \text{CH} = \text{CH}_2 \xrightarrow{\text{Benzoyl peroxide}} ?$
- (b) Arrange the following in increasing order of basicity with a suitable explanation: [4]
- (i) Ph – NH₂
- (ii) Ph₂NH
- (iii) Ph₃N
- (c) Predict the E and Z nomenclature of the following compounds: [2]
- (i) 1-bromo-1-chloroprop-1-ene
- (ii) 1-chloro-2-deuterioethene derivatives

5. Answer the following:

- (a) Write down the major products formed in the following reaction with a suitable mechanistic explanation: [8]
- $(\text{CH}_3)_2\text{CH} - \text{CH}(\text{Br}) - \text{CH}_3 \xrightarrow{\text{CH}_3\text{OK}/\text{MeOH}} ?$
- (b) What is the intermediate involved in the E1cB mechanism? [2]

- (c) Identify the products formed in the following reaction with a suitable mechanism:
[4]



6. Answer the following:

- (a) Predict the R and S nomenclature of the following chiral molecules: [3]

- (i) Glyceraldehyde derivative
- (ii) L-amino acid derivative

- (b) How do you prepare the following from phenylmagnesium iodide (PhMgI)? [6]

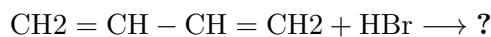
- (i) Benzenesulfonic acid (PhSO₃H)
- (ii) Phenol (PhOH)
- (iii) Benzyl alcohol (PhCH₂OH)

- (c) How do you prepare the following from ethyl acetoacetate? [5]

- (i) A 1,4-diketone
- (ii) A 1,6-diketone

7. Answer the following:

- (a) Identify the products formed in the following reaction with a suitable mechanism:
[4]



- (b) How do you prepare the following from diethyl malonate? [4]

- (i) Glutaric acid
- (ii) Dimethyl acetic acid

- (c) Discuss the S_N1 reaction mechanism with a suitable example. [4]

- (d) Distinguish between primary and secondary alcohols by the substitution method.
[2]